

Guidelines for the exam in “Advanced Signal Processing”

The exam is oral with approximately 15 minutes allocated for examining each student. In addition, for each student, approximately 5 minutes will be used for grading by the examiners.

The course consists of 3 areas, each containing four topics:

Area A topics:

1. Wiener Filtering
2. Least-Mean-Squares Adaptive Filtering
3. Recursive Least-Squares Adaptive Filtering
4. Multidimensional Fourier Transform

Area B topics:

5. Signal Spaces
6. Detection theory
7. Minimum Mean Squared Error (MMSE) and Linear MMSE estimation
8. Kalman Filtering (scalar and vector versions)

Area C topics:

9. One-step linear predictor
10. The Levinson-Durbin algorithm
11. AR-filter design by modelling
12. The Lattice FIR filter

The student will be examined in two topics belonging to two different areas. At the beginning of the exam, the student draws randomly two topics – Topic 1 and Topic 2, which belong to two different areas, A, B, or C, as listed above. The student decides the order in which the topics should be presented and discussed. As a rule of thumb, the same amount of time should be spent on each topic.

For each topic, the student initially presents his/her knowledge/understanding, using the black board for figures, equations, etc. The examiners will ask questions during the presentation regarding both conceptual and mathematical understanding.

Students are allowed to bring to the exam one sheet of notes per topic with the contents that they consider necessary to present the topics.

The course is graded on the 7-step grading scale. The examination will be conducted in English/Danish, and the examiners will be the course lecturers.

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